CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

Antifolate / nucleoside + PD-(L)1: 2 molecules stopped early, all small molecules

6 stopped trials, but 2 molecules — all small molecules: Pemetrexed and Gemcitabine. Against 5.2% for PD-(L)1 combinations without Antifolate / nucleoside and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

3.4%
6 of 177 closed trials (95% CI 1.6%–7.2%)

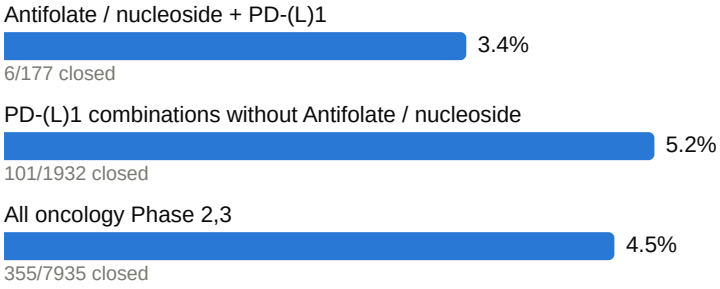
3 / 3 / 0
efficacy / safety / both (adds to 6)

29.5%
of 601 trials have closed · 424 still open or unresolved

1.2%
stopped for a biological reason within 3 years of starting (95% CI 0.4%–3.1%; 233 still at risk; 1.5% area-wide)

3 of 6 **Completed and missed the primary endpoint.** Of 121 completed trials in this segment, 6 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 3 missed and 2 met, 1 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 51 of the 109 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 308 distinct experimental drugs across 601 trials (small molecule 601, antibody 560, unknown 111, protein 51). Most tested: OXALIPLATIN (193), CAPECITABINE (188), GEMCITABINE (186), CISPLATIN (160), PACLITAXEL (144). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 6 stops came from 6 sponsor-asset programmes across 6 sponsors; the largest (Tufts Medical Center) contributed 1; removing that programme's trials from both sides leaves 2.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 19 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 14.1%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	46	2.2%	0.4%–11.3%
2018–2020	3	77	3.9%	1.3%–10.8%
2021–2024	2	54	3.7%	1.0%–12.5%

Where the stops sit

Efficacy / futility: 3, Safety: 3.
Sponsors with most stops: BerGenBio ASA (1), Tufts Medical Center (1), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (1), Nasser Hanna (1), National Cancer Institute (NCI) (1), Nathan Bahary, MD (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05469178	2	BerGenBio ASA	BEMCENTINIB, PEMBROLIZUMAB, PEMETREXED, CARBOPLATIN	2025-04	Efficacy	Lack of efficacy

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04856176	2	Tufts Medical Center	Granulocyte-Macrophage Colony-Sti…, PEMBROLIZUMAB, PEMETREXED, PACLITAXEL	2024-12	Efficacy	Futility
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.
NCT03713944	2	Nasser Hanna	CARBOPLATIN, PEMETREXED, ATEZOLIZUMAB, BEVACIZUMAB	2021-12	Safety	Extreme toxicity, thromboembolic events
NCT03451773	2	National Cancer Institute (NCI)	BINTRAFUSP ALFA, GEMCITABINE	2020-05	Safety	Study was closed after one treatment related death.
NCT03344172	2	Nathan Bahary, MD	GEMCITABINE, AVELUMAB, HYDROXYCHLOROQUINE	2019-04	Safety	Suspected Serious Adverse Events related to treatment

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 29.5% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.